Band-1 Collatz Density: Finite Trajectory + Lit Ledger (Collapse 2.07)

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Band-1 Collatz Density / Almost-Every Probe (Finite + Lit)

Honesty stamp. Every-orbit Collatz is **Band-4 OPEN**. This note does **not** claim Collatz solved. It does **not** claim a Clay-millennial solution or a Theory of Everything. **GATE-04** is locked out of scope. The box result is a **finite trajectory check**, **not** a density theorem.

Every-orbit Collatz remains OPEN. Finite verification $\neq$ almost-all theorem $\neq$ conjecture.

Abstract

Literature (analytic almost-all): Terence Tao (arXiv 2019; *Forum of Mathematics, Pi* **10** (2022)) proves that for any $f \to \infty$, $\mathrm{Col}_{\min}(N) < f(N)$ for almost all N in **logarithmic density**. Prior natural-density polynomial bounds: Korec ($\theta > \log 3/\log 4$). **2024–2026 web check:** no peer-reviewed result found that supersedes Tao 2022 on the almost-bounded almost-all statement. Preprints claiming natural-density-one convergence to 1 or full proofs are ledger-noted and **not elevated**.

Literature (computational, complementary): Barina verified convergence for all $n < 2^{68}$ (2021) and later to 2^{71} (*J. Supercomput.* 2025) — finite bounds, not density theorems.

Box finite check: every $n \in [1, 10^7]$ reaches the cycle through 1 (**0** failures; wall ≈ 7 s). Label: method certificate / finite ledger only — **orders of magnitude** below published computational records; **not** a density theorem.

Problem (collapse Band-2 item 2.07 → Band-1-style ledger)

1. Cite Tao 2019/2022 almost-all Collatz and check for subsequent **peer-reviewed density** improvements (2024–2026).
2. Certify a feasible on-box finite trajectory check for all $n \le N$.
3. Keep every-orbit conjecture, Clay, ToE, and GATE-04 off the claim surface.

Literature ledger (web check; not re-proved on box)

Year	Result	Authors	Status
1976	$\mathrm{Col}_{\min}(N) < N$ a.e. (natural dens.)	Terras; Everett	classical
1979	$\mathrm{Col}_{\min}(N) < N^{\theta}$, $\theta > 0.869$ a.e.	Allouche	classical
1994	$\mathrm{Col}_{\min}(N) \le N^{\theta}$, $\theta > \log 3/\log 4$ a.e. (natural)	Korec	classical

Year	Result	Authors	Status
2019	any $f \to \infty$: $\mathrm{Col}_{\min}(N) < f(N)$ a.e. (**log dens.**)	Tao	arXiv:1909.03562
2022	**same, published**	**Tao**, *Forum Math. Pi* 10	**standing almost-all record**
2021	all $n < 2^{68}$ converge	Barina, *J. Supercomput.*	computational
2025	all $n < 2^{71}$ converge	Barina, *J. Supercomput.* 81	computational
2025–26	FMF / Zenodo / transfer-operator density-one or "proof" claims	various preprints	**not elevated**

Canonical almost-all citation. T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum of Mathematics, Pi **10** (2022), e14 (DOI 10.1017/fmp.2022.8); arXiv:1909.03562.

Hygiene. No confirmed peer-reviewed upgrade past Tao's logarithmic-density almost-bounded theorem was found in the 2024–2026 sweep. Informal/Zenodo claims of natural-density-one reach-1 or full Collatz proofs are **not** treated as the record. Computational finite bounds (Barina 2^{71}) are complementary and do not imply density theorems.

Method (box)

1. For each $n = 1, \ldots, N$, iterate $\mathrm{Col}(x) = x/2$ (even) or $3x + 1$ (odd).
2. Stop when $x = 1$ or $x < n$ (induction: smaller starts already certified).
3. Soft step-cap safety; record max stopping-time proxy and max peak.
4. Label: **finite check** — not logarithmic/natural density.

Results

Literature status (density)

Quantity	Value
Standing almost-all analytic result	**Tao 2022** (log dens., almost bounded)
Peer-reviewed density improvement past Tao 2022	**None found** (2024–2026 web)
Every-orbit Collatz	**OPEN (Band-4)**

Box window (certified tonight)

Quantity	Value
Box N	10^7
Range	$[1, 10^7]$
Integers checked	**10{,}000{,}000**
Failures	**0**
All reach 1	**True**
Max stopping-time proxy	**401** at $n = 6{,}370{,}511$
Max peak	60,342,610,919,632 at $n = 6{,}370{,}281$
Wall (box)	≈ 7.0 s
Label	finite check $\neq$ density theorem; $\neq$ world computational record

Why Band-1 vs conjecture

- **Band-1 (this probe):** lit ledger on almost-all density; certify finite N; hygiene on preprint claims.
- **Band-4 OPEN:** every positive integer's Collatz orbit reaches 1 — **not proved here**.
- Tao almost-all $\neq$ almost-all reach 1 $\neq$ all reach 1.

Reproducibility

- Lab: `/workspace/mission-control/experiments/band1/collatz_density/`
- Run: `python3 collatz_density_finite.py`
- Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `METHOD.md`

Conclusion

Lit: Tao 2022 remains the standing peer-reviewed almost-all (log-density almost-bounded) result; **no** confirmed peer-reviewed density upgrade past Tao found in 2024–2026. **Box:** all $n \leq 10^7$ reach 1 (0 failures) — finite check only. **Every-orbit Collatz: OPEN**. No Clay / ToE claim. GATE-04 locked.